Audited Event-Time Control and Experience Admission: A Paired Mechanical Component Study

Paper B component study [1]

Abstract

Physical deadlines concern effects, not command receipts. We compare four controller lifecycles under identical public models and budgets on eight development-seen mechanical configurations. All 32 independently audited episodes satisfy the joint contract; strong batch ties feedback and replanning on success. Independent RAW separates onset from confirmation. Metadata validation admits no training rows. This simulated component establishes no VLA, chemical or agentic benefit.

1. Question and scope

Physical-time policies (Wang et al., 2026; Jia and Chen, 2026) and chemistry-twin verifiers (Darvish et al., 2026) already exist. We compare feedback and replanning against a strong same-information schedule, separating physical success from learning-data eligibility.

2. Original-event contracts

For events a, b in the same epoch, let occurrence brackets be $I_a = [\ell_a, u_a]$, $I_b = [\ell_b, u_b]$. The conservative relative interval is

$$J_{b|a} = [\ell_b - u_a,\ u_b - \ell_a].$$

A required window W is TRUE if $J \subseteq W$, FALSE if $J \cap W = \varnothing$, and UNKNOWN otherwise. Missing events are UNKNOWN. The conjunction retains FALSE/UNKNOWN, refusals, exhaustion and missing attempts in the planned denominator. A graph edge requires its minimum gap under the same brackets; success also requires its physical predicates.

The fixture contains three contact-free MuJoCo rotors. B's first actual $q = 2$ crossing is an immutable original reference. A must stop within $[1.8, D]$ seconds of B; C within $D \pm .15$ seconds, with a 40 ms graph gap. B's required motion and drive profile are checked over its original six-second interval. The RAW endpoint is the first stop with prior movement and 21 samples at speed $\leq .1$ spanning 200 ms on the 10 ms grid. Occurrence differs from confirmation; sampled predicates do not prove continuity between samples.

3. Public control and paid decisions

Actors receive lagged position, velocity, torque, capture/arrival times, reservations and ACKs. Same-tick capture is pre-ACK; effect evidence requires capture $\geq$ACK+10 ms. ACK is not stopping. Every actor may conditionally refine the *same* B event with the unfitted public model. Coarse 50 ms observation is not an information lower bound; compatibility does not certify physical support.

A capacity-one bus services A/C and B's reservation. The first A ACK leaves A powered; public hypotheses are none/first-drop. Accepted requests cannot be cancelled/refunded. Caps are two requests per rotor, four total, and 1,000 candidates per decision/10,000 per epoch including resource rejections. Each decision costs 20 ms, two native ticks; old commitments and physics continue. Python compute otherwise pauses physics: this charge is not LLM wall time.

Four distinct lifecycles. Strong batch commits a complete joint schedule with blind redundancy. Guarded HSM uses a safe feedback retry slot or the same blind fallback. EDF replans unissued requests with inherited deadlines. Interval MPC commits one prefix and recedes on public changes or a 100 ms cadence. The ledger charges at emission; paid replacements with stale request or calendar history are discarded. Failed bounded search does not prove infeasibility.

4. Paired evidence and attribution

Graph order, B service 50/450 ms and $D = 2.8/3.1$ seconds yield eight fixed configurations. D changes A's upper bound and C's window together. One episode per method/configuration gives 32 episodes, eight per method, with unique identities and the same initial state. All configu-

[1] Simulation and evidence-interface evaluation. Correspondence to: Paper B component report.

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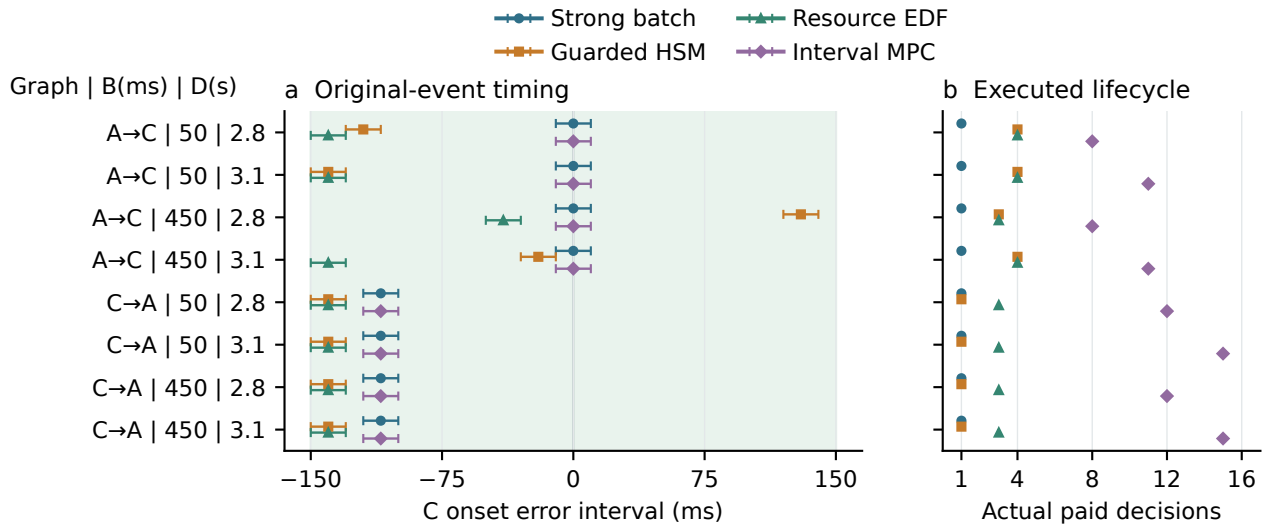


Figure 1. All 32 audited development episodes. Left: C onset-error intervals relative to the original B event, with the allowed ±150 ms window shaded. Right: actual charged decision counts. Intervals, not midpoints, determine certification. Different workloads do not establish an agentic or statistical advantage.

rations were previously seen; no random-seed, IID, heldout or OOD claim follows.

Table 1. Actual audited episodes: eight per method. T/F/U is the joint physical contract. Charge is virtual decision time.

Method	T/F/U	Dec.	Proposals	Charge (ms)
Strong batch	8/0/0	1	282–370	20
Guarded HSM	8/0/0	1–4	286–901	20–80
Resource EDF	8/0/0	3–4	284–1197	60–80
Interval MPC	8/0/0	8–15	1548–3131	160–300

All methods are 8/8 joint TRUE (Table 1), with zero FALSE/UNKNOWN and A2/C1 requests throughout. Strong batch already explains all eight configurations; no feedback, replanning or learning benefit is demonstrated. Workloads are descriptive under shared caps. Maximum actual A/C queue waits are 0/0/20/10 ms for batch/HSM/EDF/MPC; B reservation wait is separate.

The 64 scalar software checks (8 configurations×4 methods×2 fault hypotheses) all pass, including batch; they are not native episodes. The separate native prerequisite is TRUE, with one primary and one diagnostic re-integration, 1,201 samples/1,200 steps each. It is not a paired32 outcome. The 32 controlled episodes contain 38,432 RAW samples/38,400 steps. Separately, 32 diagnostic command re-integrations add 38,432 samples/38,400 steps with zero state/time error. Replays are not new primary trials.

5. Evidence admission is not learning

The replay interface preserves four legacy calibration, sixteen test and eight development records: 28 metadata records, not new robot tests. Its 852 image-action candidates have 832 UNKNOWN policy-image delivery times and 20 REJECTED tail frames with no future submission. Applied control or post-hoc effect is not a future submission label. Thirty-four CPU schema/cross-field checks and clean public metadata replay pass; student tensors and DP/offline-RL/OPD training rows remain zero. This metadata gate postdates the legacy outcomes.

The actors are rules, not neural teachers; aggregation without student rollouts is not on-policy distillation. Learning requires causal observations, learner-state labels, matched budgets and unseen tests. Loaded stirring, flow, heating and chemical endpoints need separate validation. This study uses no GPU, LLM calls or weight updates.

References

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